

Mobile-Optimized AI-Based Thrips Detection and Infestation Severity Assessment in Onion Crops Using Object Detection Models

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Onion (*Allium cepa*) cultivation is a cornerstone of Sri Lankan agriculture, yet crop yields are severely threatened by Thrips (*Thysanoptera*), one of the most destructive pests affecting onion crops worldwide. These minute, sap-sucking insects cause characteristic silvery-white feeding streaks on leaves, leading to reduced photosynthetic capacity, stunted growth, and yield losses of 20–40% under heavy infestations. Current pest monitoring methods rely on labour-intensive manual scouting, sticky traps, and periodic laboratory analysis—approaches that are time-consuming, inconsistent, and inaccessible to the majority of Sri Lanka's smallholder onion farmers.

This research proposes the design, development, and validation of a mobile-optimized, AI-based pest detection system for real-time identification of Thrips and assessment of infestation severity in onion crops. The system employs state-of-the-art object detection frameworks—YOLOv8 and Faster R-CNN—trained on a locally collected dataset of 2,500+ annotated pest images from FCRDI demonstration plots and partner farms. AutoML optimizes YOLOv8 anchor boxes and hyperparameters for small-object detection, while TinyML techniques

enable deployment with severity heatmaps quantifying infestation levels for Integrated Pest Management (IPM) thresholds.

The expected outcome is a high-accuracy pest detection model capable of identifying Thrips from mobile-captured field images, classifying infestation severity (Low / Medium / High), and providing IPM recommendations and preventive actions. This system directly addresses the critical gap in automated, image-driven pest identification for Sri Lankan onion agriculture, empowering farmers with real-time, field-level diagnostic capability.

Keywords: Thrips Detection, YOLOv8, Faster R-CNN, Object Detection, Small-Object Detection, Onion Pest Management, TinyML, AutoML, IPM

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
YOLO	You Only Look Once
R-CNN	Region-based Convolutional Neural Network
CNN	Convolutional Neural Network
mAP	Mean Average Precision

IoU	Intersection over Union
FPN	Feature Pyramid Network
NMS	Non-Maximum Suppression
AutoML	Automated Machine Learning
TinyML	Tiny Machine Learning
IPM	Integrated Pest Management
FCRDI	Field Crops Research and Development Institute
DOA	Department of Agriculture
FAO	Food and Agriculture Organization
SDG	Sustainable Development Goal
TFLite	TensorFlow Lite
INT8	8-bit Integer Quantization
API	Application Programming Interface
UI	User Interface
WBS	Work Breakdown Structure

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Appendix A	Thrips Infestation Severity Classification Guide
Appendix B	Image Collection and Annotation Protocol
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1. INTRODUCTION

1.1 Background and Context

Onion (*Allium cepa*) cultivation is one of the most economically significant agricultural activities in Sri Lanka, contributing substantially to both national food security and the livelihoods of rural farming communities. According to the Department of Agriculture Sri Lanka, onion is cultivated across multiple agro-ecological zones, with the Maha Illuppallama region being a primary production hub. Despite its importance, onion farmers face persistent and multifaceted challenges that severely impact crop health and economic returns.

The Real-World Problem:

Among the many threats to onion cultivation in Sri Lanka, Thrips (*Thysanoptera*) represent one of the most damaging and difficult-to-manage pest populations. These minute, slender-bodied insects (typically 1–2 mm in length) feed by rasping the leaf surface and sucking the exuded sap, producing characteristic silvery-white streaks, curling, and eventual leaf desiccation [1]. Under favourable warm and dry conditions—prevalent in Sri Lanka's onion-growing Dry Zone—thrips populations can explode rapidly, causing yield reductions of 20–40% [2]. Early detection is critical because thrips are extremely small and often hide within leaf folds, making manual identification challenging until visible damage has already progressed significantly. Additionally, fungal diseases such as purple blotch (*Alternaria porri*) and nutrient deficiencies in nitrogen, potassium, and phosphorus further compound crop vulnerability [3].

Who Is Affected and Why It Matters:

Sri Lanka has an estimated 150,000+ smallholder onion farming households. These farmers typically have limited formal education and rely on subjective visual inspection and manual scouting for pest monitoring—methods that are inconsistent, time-consuming, and prone to error, especially for tiny pests like thrips. By the time damage symptoms become clearly visible, significant yield loss has already occurred. The Food and Agriculture Organization (FAO) reports that post-harvest losses in onion crops in developing countries can reach

30–40%, much of which is attributable to preventable pest damage [4].

Statistical Evidence:

- Sri Lanka imports approximately 200,000 metric tons of onions annually, highlighting a significant domestic production shortfall [5].
- Studies indicate that proper pest management could increase onion yields by 25–35% in the South Asian context [6].
- Only 12% of Sri Lankan farming households have access to digitalized agricultural advisory services [3].
- Thrips are reported as the primary insect pest of onion in all major growing regions of Sri Lanka [1].

Analysis of Existing Solutions and Their Limitations:

Current pest monitoring approaches in Sri Lankan onion cultivation rely primarily on manual scouting, sticky trap counts, and laboratory-based identification. While global agri-tech platforms such as Plant.id and Plantix offer CNN-based crop health diagnostics, they are designed for disease classification rather than pest detection and lack the object detection capability required to localize and count small insects like thrips in field images. Existing object detection research using YOLOv5 and Faster R-CNN has demonstrated strong performance on insect datasets, but onion-specific pest detection models remain extremely scarce [7]. Furthermore, no existing system provides severity-graded infestation assessment linked to IPM threshold-based recommendations for Sri Lankan conditions.

The Proposed Solution:

This research proposes a mobile-optimized, AI-based pest detection system for real-time identification of Thrips in onion crops using object detection models. The system employs YOLOv8 and Faster R-CNN frameworks trained on a locally collected and expert-annotated dataset of 2,500+ pest images from FCRDI demonstration plots and partner farms. AutoML optimizes anchor box configurations and hyperparameters for small-object detection, while TinyML techniques enable on-device deployment with severity heatmaps. The overall platform comprises four integrated components:

1. **Component 1: CNN-Based Purple Blotch Disease Detection**

- (Member 1 – IT22054890)
2. **Component 2:** Pest Detection & Identification using YOLOv8/Faster R-CNN (Member 2 – IT22226464) (*This component*)
 3. **Component 3:** Multilingual Agro-Advisory Chatbot – Sinhala/English/Singlish (Member 3 – IT22087256)
 4. **Component 4:** Image-Based Nutrient Deficiency Detection (Member 4 – IT22142528)

SDG Alignment:

This project directly contributes to:

- **SDG 2 (Zero Hunger):** By enabling early pest detection that protects crop yields and reduces food production losses.
- **SDG 12 (Responsible Consumption and Production):** By reducing excessive and inappropriate pesticide application through data-driven, severity-based IPM recommendations.
- **SDG 13 (Climate Action):** By promoting sustainable pest management practices that minimize chemical inputs and environmental impact.

Research Cluster Alignment:

This research is aligned with the Autonomous Intelligent Machines and Systems (AIMS) research cluster and the Agri-Technology industry vertical, focusing on intelligent perception, decision-making, and human-centric AI solutions.

1.2 Literature Review

A comprehensive literature review was conducted examining recent scholarly works in object detection for pest identification, small-object detection techniques, agricultural pest monitoring systems, and mobile-optimized model deployment strategies. The following key findings emerged:

1. Object Detection Frameworks for Pest Identification:

Redmon et al. (2016) introduced YOLO (You Only Look Once), a real-time object detection framework that processes entire images in a single forward pass, achieving high speed and reasonable accuracy [8]. The

YOLOv5 and YOLOv8 iterations significantly improved small-object detection through enhanced Feature Pyramid Networks (FPN), better anchor-free detection heads, and mosaic augmentation. Ren et al. (2015) proposed Faster R-CNN, a two-stage detector using Region Proposal Networks (RPNs) for precise object localization, achieving higher accuracy at the cost of slower inference [9]. Studies applying these frameworks to insect detection report mAP scores above 85% on diverse pest datasets.

2. Small-Object Detection Challenges in Entomology:

Thrips present a unique challenge for object detection due to their extremely small body size (1–2 mm), translucent wings, and tendency to hide within leaf folds and growing points. Liu et al. (2021) reviewed small-object detection techniques, finding that multi-scale feature fusion, anchor box optimization, and high-resolution input images are critical for detecting objects occupying fewer than 32×32 pixels in the image [10]. Data augmentation strategies including random crop, mosaic, and copy-paste augmentation have been shown to improve small-object detection by 5–10% mAP.

3. AI-Based Insect Pest Detection in Agriculture:

Li et al. (2022) developed a lightweight insect detection model using YOLOv5 for rice pest identification, achieving 91.2% mAP on a field-collected dataset with aggressive data augmentation [11]. Their work highlighted class imbalance challenges, as pest images are inherently rare compared to background images, and proposed focal loss and oversampling as effective mitigation strategies. However, onion-specific pest detection remains largely unexplored.

4. Damage Pattern Recognition for Infestation Assessment:

Beyond detecting individual insects, several studies have explored using CNN-based classification to assess crop damage severity from pest feeding patterns. Fuentes et al. (2017) demonstrated a multi-task detection approach combining pest identification with damage classification in tomato crops using Faster R-CNN [12]. Applying damage-pattern analysis alongside direct pest detection provides more robust infestation assessment, particularly when pest individuals are not visible but damage symptoms are evident.

5. AutoML for Object Detection Optimization:

Wang et al. (2021) introduced FLAML, a fast and lightweight AutoML framework for efficient hyperparameter optimization [13]. When applied to object detection, AutoML can systematically optimize anchor box configurations, learning rates, augmentation strategies, and architectural choices (e.g., backbone selection, neck design) to maximize mAP on domain-specific datasets without exhaustive manual experimentation.

6. TinyML for Edge-Deployed Pest Detection:

Banbury et al. (2021) surveyed TinyML techniques for deploying deep learning models on resource-constrained devices [14]. For object detection specifically, INT8 quantization and layer pruning can reduce YOLOv8 model sizes by 3–4× while maintaining >90% of original mAP, enabling real-time inference on mid-range smartphones. This approach is critical for agricultural applications in rural areas with limited internet connectivity.

7. Existing Pest Monitoring Systems and Their Limitations:

Current agrimeteorology tools provide weather-based pest risk alerts (e.g., temperature and humidity thresholds for thrips population surges) rather than image-driven identification. Commercial crop health apps like Plantix and Agrio focus primarily on disease classification and lack dedicated object detection modules for insect identification. Farmer reliance on manual scouting and sticky trap counts underscores the absence of integrated mobile detectors that estimate infestation levels from field images [7]. The Sri Lankan DOA provides static pest management guidelines but no interactive, real-time pest identification tools.

8. RetinaNet and Focal Loss for Imbalanced Detection:

Lin et al. (2017) introduced RetinaNet with focal loss, specifically designed to address the extreme foreground-background class imbalance inherent in object detection [15]. For pest detection, where pest instances may occupy <1% of the image area, focal loss down-weights the contribution of easy negative examples, focusing training on hard positives—a critical technique for detecting rare, small-object pest instances.

1.3 Research Gap

The comprehensive literature review reveals several critical gaps that this research aims to address:

1. No Onion-Specific Thrips Detection Model Using Object Detection:

While object detection frameworks like YOLO and Faster R-CNN have been successfully applied to pest detection in rice, tomato, and cotton crops, no model has been specifically trained and validated for detecting Thrips (*Thysanoptera*) in Sri Lankan onion crops. Existing research uses generic insect datasets that do not capture the specific morphology, colour, and behaviour of thrips on onion leaves under tropical field conditions [7].

2. Small-Object Detection Challenge Unaddressed for Thrips:

Thrips are among the smallest agricultural pests (1–2 mm body length), making them particularly challenging for standard object detection models trained at typical input resolutions. No study has specifically optimized anchor box configurations, input resolution, and feature pyramid networks for thrips-sized objects in onion crop images.

3. Absence of Severity-Graded Infestation Assessment:

Current AI pest detection systems typically provide binary outputs (pest present / absent) or simple pest counts. No existing system provides severity-graded infestation assessment (Low / Medium / High) linked to Integrated Pest Management (IPM) threshold-based action recommendations specifically for onion thrips in Sri Lankan contexts.

4. No TinyML-Optimized Pest Detection for Offline Field Use:

Existing pest detection research deploys models on cloud servers, requiring internet connectivity for inference. No pest detection model has been specifically optimized using TinyML techniques to enable offline, on-device inference on smartphones commonly used by Sri Lankan farmers.

5. No Integration with Multilingual Advisory Systems:

Current pest detection tools provide diagnostic outputs in technical English without farmer-friendly IPM recommendations. No existing system integrates object detection-based pest identification with a multilingual advisory pipeline capable of explaining pest results and recommending control measures in Sinhala or Singlish.

Evidence-Based Justification for Novelty:

Feature	Plant.id	Plantix	Agrio	Generic YOLO Pest Models	Proposed System
Onion-Specific Thrips Detection	x	x	x	x	✓
Object Detection (not just classification)	x	x	Partial	✓	✓
Small-Object Optimized (anchor tuning)	x	x	x	x	✓
Severity-Graded Infestation (Low/Med/High)	x	x	x	x	✓
IPM Threshold-Based Recommendations	x	x	x	x	✓
TinyML Offline Deployment	x	x	x	x	✓
Severity Heatmap Visualization	x	x	x	x	✓
Multilingual Advisory Integration	x	x	x	x	✓

Table 1: Comparison of Existing Pest Detection Systems

2. OBJECTIVE

2.1 Main Objective

To design, develop, and validate a mobile-optimized, image-based pest detection system for real-time identification of Thrips (*Thysanoptera*) in onion crops using object detection models (YOLOv8 / Faster R-CNN), achieving >85% mAP on a locally collected dataset and deployable for offline inference on smartphones, with severity-graded infestation assessment and IPM-based recommendations validated through field testing with FCRDI entomologists.

2.2 Specific Objectives

1. **Collect and Annotate a Locally Representative Pest Image Dataset:** Collect 2,500+ pest images from FCRDI demonstration plots and 15–20 partner commercial farms, including thrips damage and insects (1,500+), healthy plants (600+), and mixed pest damage (400+), with expert entomologist labeling and bounding box annotation.
2. **Implement and Compare Object Detection Architectures:** Evaluate YOLOv8 (anchor-free, single-stage) and Faster R-CNN (two-stage) architectures for thrips detection, with specialized feature extraction based on pest morphology and damage patterns, addressing the small-object detection challenge through optimized input resolution and multi-scale feature fusion.
3. **Optimize Small-Object Detection Using AutoML:** Deploy AutoML to systematically search across anchor box configurations, backbone architectures, hyperparameters, and augmentation strategies optimized for thrips-sized objects, maximizing mAP on the local pest dataset.
4. **Develop Severity-Graded Infestation Assessment:** Implement a multi-level infestation severity classification system (Low / Medium / High) based on detected pest counts, spatial distribution, and damage pattern analysis, linked to IPM threshold-based action recommendations from FCRDI guidelines.
5. **Optimize for Mobile Deployment Using TinyML Techniques:**

Apply INT8 quantization and model pruning to reduce the best-performing model for deployment on resource-constrained Android devices, enabling offline, real-time inference with severity heatmaps quantifying infestation levels.

6. **Validate Through Field Testing with Entomologists:** Conduct field validation with FCRDI entomologists and onion farmers, measuring detection accuracy (mAP), classification precision, infestation severity agreement with expert assessment, and practical usability.
-

3. METHODOLOGY

Research Approach

This research follows a **Design Science Research (DSR)** methodology, which emphasizes the iterative design, development, and evaluation of an IT artifact (the pest detection model) to solve an identified real-world problem. The DSR framework guides the process through problem identification, solution design, development, demonstration, and evaluation cycles [16].

Technical Methods and Algorithms

1. Data Collection and Annotation:

- **Primary Source:** FCRDI onion demonstration plots (Maha Illuppallama) and 15–20 partner commercial farms
- **Image Capture:** High-resolution images of onion leaves and plants showing pest presence or damage, captured using smartphone cameras under natural field conditions
- **Categories:**
 - Thrips damage and insects (1,500+)
 - Healthy plants (600+)
 - Mixed pest damage (400+)

- **Annotation:** Bounding box annotation using LabelImg/CVAT with pest species labels and damage severity indicators. FCRDI entomologists verify pest identification and severity classification
- **Data Augmentation:** Mosaic augmentation, random crop, horizontal/vertical flip, brightness/contrast variation, copy-paste augmentation for rare pest instances, scale jittering for small-object robustness

2. Object Detection Model Architectures:

Two primary architectures will be evaluated:

- **YOLOv8 (Ultralytics):** Single-stage, anchor-free detector with CSPDarknet backbone, C2f modules, and multi-scale detection heads (P3/P4/P5). Advantages include real-time inference speed and state-of-the-art small-object detection with mosaic augmentation.
 - Input resolution: 640×640 (upscaled to 1280×1280 for small-object optimization)
 - Loss: CloU loss + DFL (Distribution Focal Loss) + BCE
 - Training: SGD optimizer, cosine LR scheduling, 300 epochs with early stopping
- **Faster R-CNN (Detectron2):** Two-stage detector with ResNet-50-FPN backbone, Region Proposal Network (RPN), and ROI Align. Advantages include higher localization precision for small objects.
 - Input resolution: 800×1333 (short edge 800)
 - Loss: Smooth L1 + Cross-entropy
 - Training: SGD with momentum, step LR scheduling, 36 epochs (3× schedule)

3. Small-Object Detection Optimization:

- High-resolution input processing (1280×1280 for YOLOv8)
- Feature Pyramid Network (FPN) with additional P2 detection head for smallest objects
- Anchor box optimization using k-means clustering on thrips bounding box dimensions
- Test-time augmentation (TTA) with multi-scale inference
- Sliced inference (SAHI) for processing high-resolution field images

4. AutoML Hyperparameter Optimization:

AutoML will automate the search across:

- Backbone selection (CSPDarknet, ResNet-50, ResNet-101, EfficientNet)
- Input resolution, batch size, learning rate, weight decay
- Augmentation strategy combinations and intensities
- NMS threshold, confidence threshold optimization
- Evaluation using 5-fold cross-validation with mAP@0.5 and mAP@0.5:0.95 as selection criteria

5. Infestation Severity Assessment:

- **Low:** 1–5 thrips detected per image region; minimal visible damage
- **Medium:** 6–15 thrips detected; moderate silvery-white streaking visible
- **High:** >15 thrips detected; extensive leaf damage, curling, desiccation
- Severity heatmap generated using detection density and spatial distribution analysis
- IPM recommendations mapped to severity levels based on FCRDI pest management protocols

6. TinyML Optimization:

The best-performing model undergoes:

- **Post-Training INT8 Quantization:** Weights and activations quantized from FP32 to INT8
- **Channel Pruning:** Removal of low-magnitude convolutional filters followed by fine-tuning
- **Target:** Inference latency <1 second on mid-range Android device; acceptable mAP degradation <3%

Tools and Platforms

Tool/ Platform	Purpose	Justification
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Ultralytics YOLOv8	Primary object detection framework	State-of-the-art anchor-free detector with excellent small-object support
Detectron2	Faster R-CNN implementation	Facebook AI's modular detection library with FPN support
PyTorch	Deep learning framework	Flexible, strong community support for detection research
CVAT / Labellmg	Image annotation	Open-source, supports bounding box annotation in YOLO/COCO format
SAHI	Sliced inference for high-res images	Enables detection of small objects in full-resolution field images
Flutter	Mobile application development	Cross-platform (Android/iOS), single codebase
Flask/ FastAPI	API backend development	Lightweight, Python-native, ideal for ML model serving
Firebase	Real-time database & authentication	Scalable, free tier available, mobile-optimized
TensorFlow Lite	Mobile model deployment	Native Android support, INT8 quantization tools
Git/GitHub	Version control	Industry standard for collaborative development

Data Collection and Ethical Considerations

Data Sources:

- **FCRDI Demonstration Plots:** Primary source of pest images collected under natural field conditions
- **Partner Commercial Farms:** 15–20 farms providing diverse infestation scenarios across growth stages and environmental

conditions

- **Expert Labeling:** FCRDI entomologists identify pest species, verify bounding box annotations, and assign severity levels

Data Format: 2,500+ images across 3 categories:

- Thrips damage and insects (1,500+)
- Healthy plants (600+)
- Mixed pest damage (400+)

Annotation Protocol:

- Primary bounding box annotation by trained student researchers following FCRDI pest identification guide
- Verification and correction by FCRDI entomologists
- Inter-annotator agreement (IoU-based) computed on 10% sample subset
- Annotations stored in YOLO format (class, x_center, y_center, width, height) and COCO JSON format

Ethical Considerations:

- This project requires ethical clearance (submitted to Research Ethics Committee)
- Farmer consent forms obtained for all on-farm photography
- All farmer data anonymized; no personally identifiable information stored
- Image metadata stripped of GPS coordinates before storage
- Data management plan with confidentiality protocols
- Compliance with Sri Lankan data protection regulations

Validation Strategy and Performance Metrics

Metric	Target	Method
mAP@0.5	≥ 85%	5-fold cross-validation on test set
mAP@0.5:0.95	≥ 65%	Evaluation across IoU thresholds
Precision (Thrips class)	≥ 85%	Per-class precision analysis

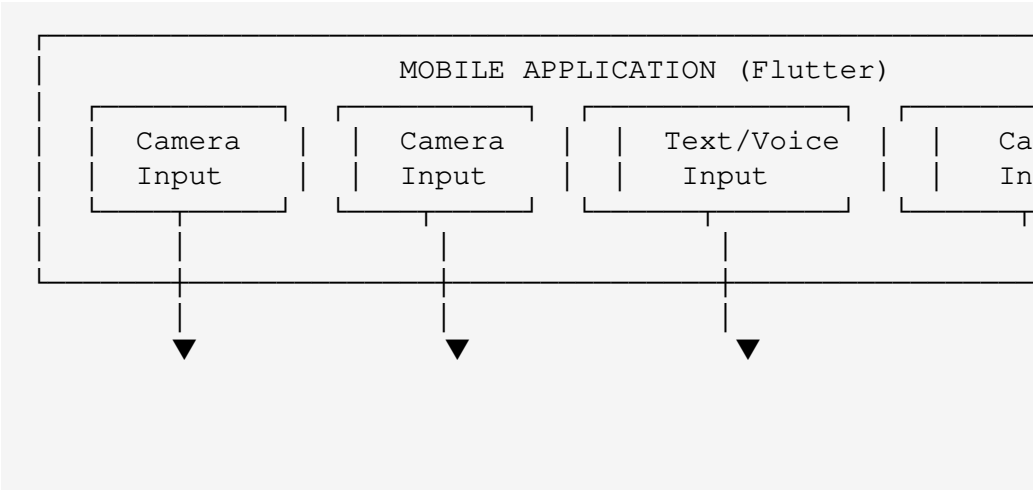
Recall (Thrips class)	≥ 80%	Per-class recall analysis
Inference Latency	< 1 second	Benchmarking on mid-range Android device (Snapdragon 665)
mAP Drop Post-Quantization	< 3%	Comparison of FP32 vs. INT8 model mAP
Severity Classification Agreement	≥ 80%	Comparison with entomologist assessment on 200 field samples
False Positive Rate	< 15%	Measured on healthy plant images

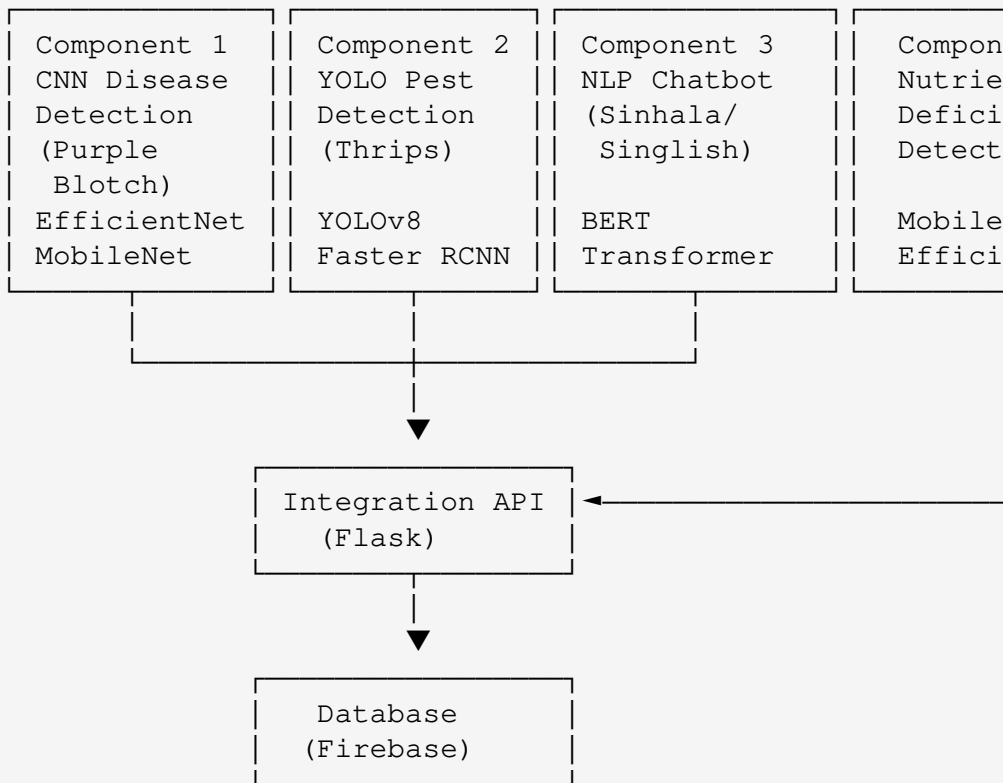
4. HIGH-LEVEL SYSTEM ARCHITECTURE

Overview of the Proposed Solution

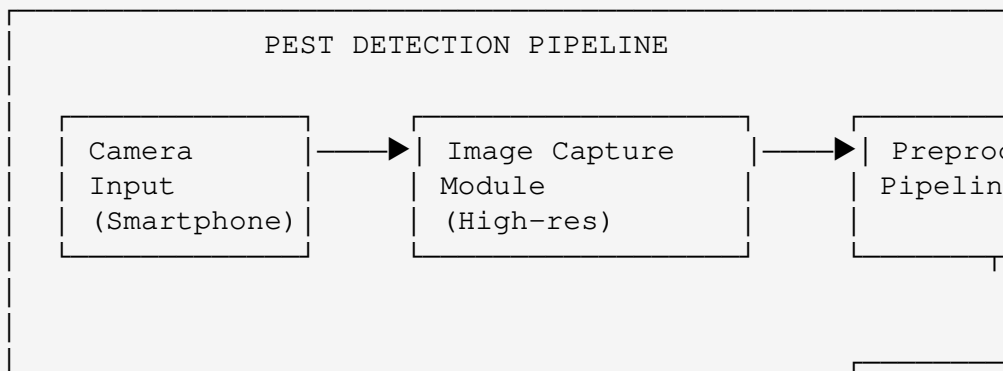
The Automated Diagnostic Framework for Onion Crop Health Monitoring is a mobile-first AI platform comprising four integrated components. This section describes the overall system and highlights the Pest Detection & Identification (Component 2) within the larger architecture.

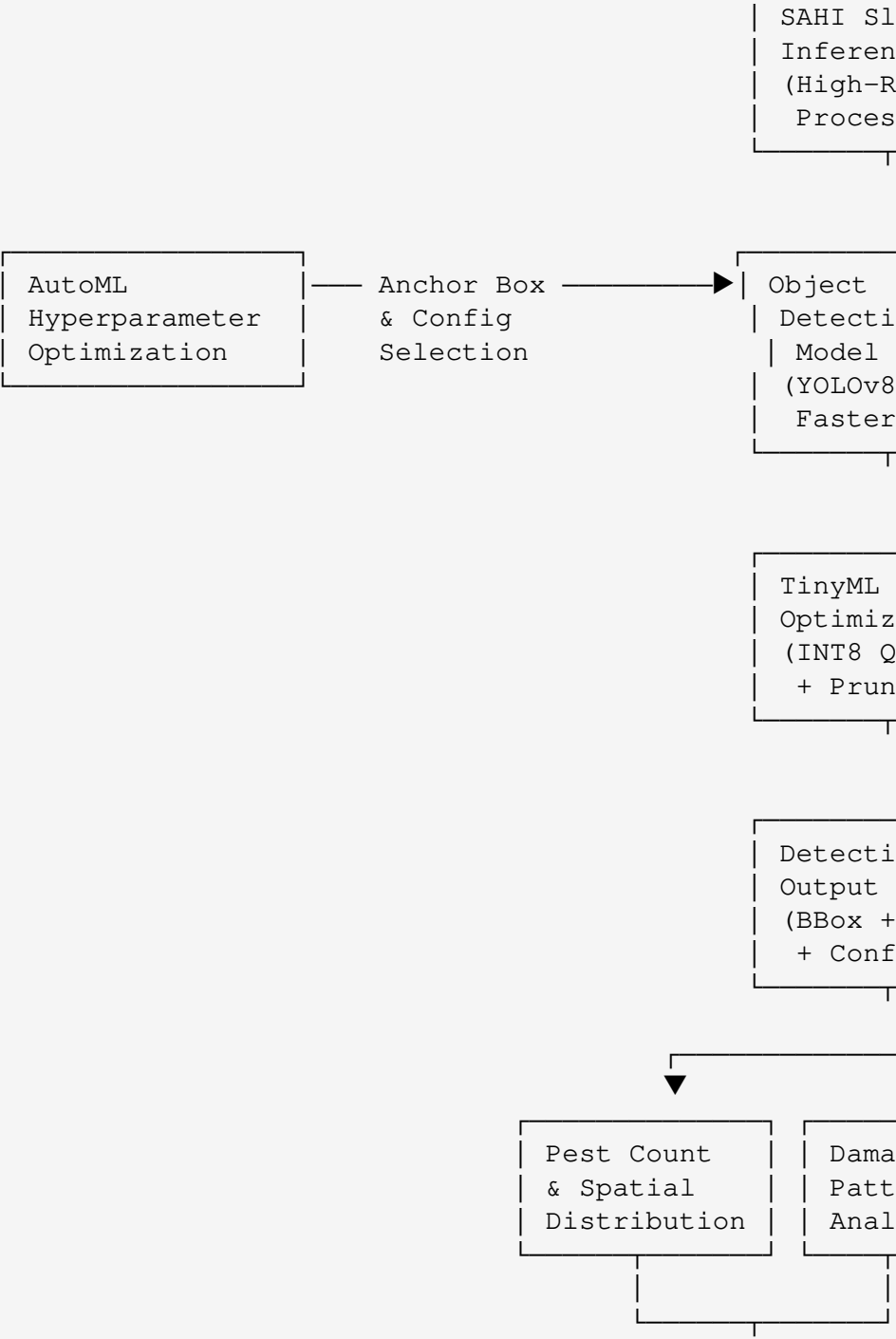
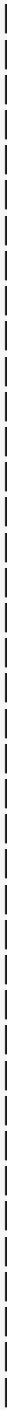
System Diagram of Components

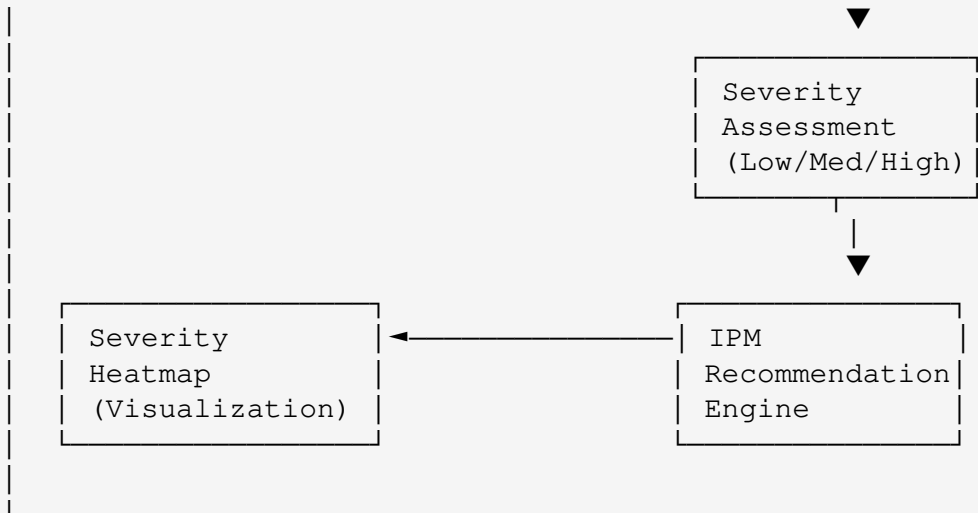




Detailed Pest Detection Component Architecture (Component 2)







Component Interactions

The pest detection module (Component 2) interacts with other system components through well-defined REST APIs:

- **Component 2 → Component 3:** When the object detection model identifies thrips, it sends the pest type, infestation severity (Low/Medium/High), detected pest count, and severity heatmap to the chatbot API. The chatbot generates IPM recommendations in the farmer's preferred language (Sinhala/English/Singlish).
- **Component 2 → Component 1:** Pest damage information is shared with the disease detection module to help differentiate pest damage patterns (silvery streaks from thrips) from disease lesions (purple-brown spots from *Alternaria porri*), reducing misclassification.
- **Component 2 → Integration API:** Detection results, pest counts, severity classifications, timestamps, and environmental metadata are stored via the Flask API for historical tracking, seasonal trend analysis, and model improvement.

Data Flow

1. Farmer captures an image of the onion leaf/plant using the mobile app camera
2. Image is preprocessed: resized, normalized, and optionally sliced (SAHI) for high-resolution processing
3. The object detection model performs inference, generating bounding boxes around detected thrips and damage areas
4. Pest count and spatial distribution are analyzed to determine infestation severity (Low/Medium/High)
5. Severity heatmap is generated showing infestation density across the image
6. IPM recommendation engine maps severity to appropriate control measures from FCRDI protocols
7. Results are displayed to the farmer and sent to the chatbot for multilingual explanation

Scalability and Security Considerations

- **Scalability:** The TinyML-optimized model runs on-device, eliminating server-side scaling concerns for inference. The AutoML framework creates an extensible architecture search pipeline. The training pipeline supports incremental learning as new pest data is collected across seasons.
 - **Security:** All API communications use HTTPS encryption. Farmer image data is anonymized at the point of collection with GPS metadata stripped. Firebase authentication ensures secure access control. The system complies with Sri Lankan data protection requirements.
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5. USER REQUIREMENTS

Stakeholder Identification

Stakeholder	Role	Primary Need
Onion Farmers	End Users	Rapid, accurate pest identification with treatment advice
FCRDI Field Officers	Advisory Support	Validate detection accuracy, monitor pest population trends
Entomologists	Domain Experts	Ensure detection accuracy matches expert identification
System Administrators	Technical Support	Manage model updates and performance monitoring

Functional Requirements

ID	Requirement	Priority
FR-01	The system shall accept onion leaf/plant images captured via smartphone camera	High
FR-02	The system shall detect and localize thrips in captured images using bounding boxes	High
FR-03	The system shall classify detected pests by species (Thrips) with confidence scores	High
FR-04	The system shall assess infestation severity as Low, Medium, or High based on pest count and damage extent	High
FR-05	The system shall generate severity heatmaps showing spatial distribution of infestation	High
FR-06	The system shall provide IPM-based recommendations and preventive actions based on severity level	High
FR-07	The system shall perform inference on-device for offline capability	High
FR-08	The system shall send detection results to the chatbot API for multilingual explanation	Medium

FR-09	The system shall log detection results for historical tracking and seasonal analysis (anonymized)	Medium
FR-10	The system shall support high-resolution image processing via sliced inference (SAHI)	Medium
FR-11	The system shall display detection confidence alongside bounding box results	Medium
FR-12	The system shall support model updates via over-the-air (OTA) distribution	Low

Table 2: Functional Requirements

Non-Functional Requirements

ID	Requirement	Category	Target
NFR-01	Detection mAP@0.5 shall exceed 85% on test set	Accuracy	$\geq 85\%$
NFR-02	Inference latency shall be under 1 second on target device	Performance	$< 1s$
NFR-03	Optimized model size shall be suitable for mobile deployment	Efficiency	$\leq 30 \text{ MB}$
NFR-04	mAP degradation after quantization shall be under 3%	Accuracy	$< 3\%$
NFR-05	The UI shall be intuitive for users with limited digital literacy	Usability	$SUS \geq 70$
NFR-06	All API communications shall use HTTPS encryption	Security	TLS 1.2+
NFR-07	Farmer image data shall be anonymized and stripped of GPS metadata	Privacy	Compliance

NFR-08	The system shall function offline for core pest detection	Availability	Offline mode
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Table 3: Non-Functional Requirements

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

Primary Market: Sri Lanka's 150,000+ onion farming households concentrated in the Dry Zone (Maha Illuppallama, Matale, Dambulla, Jaffna).

Customer Persona:

- **Name:** Kamal (representative farmer)
- **Age:** 35–55 years
- **Education:** O/L or below
- **Language:** Sinhala, Singlish
- **Technology:** Owns a basic Android smartphone, limited data plan
- **Pain Point:** Cannot detect thrips early enough due to their tiny size; relies on visible damage symptoms by which time significant yield loss has occurred; often applies pesticides excessively or ineffectively without proper infestation assessment

Value Proposition

A single mobile application that allows farmers to photograph their onion plants, receive instant AI-powered pest detection with severity grading and pest counts, and get specific IPM-based treatment recommendations—all working offline without internet connectivity, replacing the need for labour-intensive manual scouting and delayed expert identification.

Revenue Model

The system follows a **Freemium** approach:

Tier	Features	Price
Free	Basic pest detection, pest identified, general IPM advice	Free
Premium	Severity-graded assessment, heatmap visualization, detailed IPM recommendations, historical tracking, chatbot integration	LKR 300/month per farmer
Enterprise	Bulk licensing for agricultural cooperatives, white-label options, analytics dashboard, regional pest outbreak monitoring	Custom pricing

Cost Estimation

Item	Estimated Annual Cost
Cloud infrastructure (GCP – training only)	LKR 96,000
App Store + Play Store hosting	LKR 38,454
Data storage (Firebase + GCS)	LKR 12,000
GPU compute for model retraining	LKR 48,000
Marketing and farmer outreach	LKR 50,000
Total	LKR 244,454

Competitive Advantage

1. **First onion-specific thrips detection model** trained on Sri Lankan field images
2. **Object detection (not just classification)** with bounding box localization and pest counting

- 3. **Small-object optimized** with AutoML-tuned anchor boxes and SAHI integration
- 4. **Severity-graded infestation assessment** with IPM threshold-based recommendations
- 5. **Severity heatmap visualization** showing spatial infestation patterns
- 6. **TinyML offline deployment** for rural areas with limited connectivity
- 7. **FCRDI-validated** pest identification and IPM recommendations
- 8. **Integration with multilingual chatbot** for Sinhala/Singlish advisory

Intellectual Property Considerations

- The trained object detection models and locally collected onion pest dataset constitute protectable intellectual property
- The AutoML-optimized small-object detection pipeline and severity assessment algorithm may be published as research contributions
- Open-source components (YOLOv8, Detectron2, Flutter) are used under their respective licenses (AGPL-3.0, Apache 2.0, BSD)

7. BUDGET AND JUSTIFICATION

Item	Description	Cost (LKR)	Occurrence	Justification
Cloud GPU Instance	GCP Compute Engine (T4 GPU) for model training	8,000	Monthly	GPU-accelerated training required for YOLOv8/Faster R-CNN with high-resolution inputs

Firebase (Realtime DB + Auth)	Database hosting and user authentication	0 (Free tier)	Monthly	Free tier sufficient for initial deployment (1GB storage, 10K auth/ month)
Play Store Developer Account	Publishing the mobile app on Google Play	7,754	One-Time	Required for Android app distribution
App Store Developer Account	Publishing on Apple App Store	30,700	Annual	Required for iOS app distribution
Cloud Storage (GCS)	Storing training images, annotations, model weights	0.023 USD/ GB	Monthly	Cost-effective cloud storage for 2,500+ images and model artifacts
Domain Name	Custom domain for API endpoints	3,000	Annual	Professional branding and consistent API access
Field Visit Transportation	Travel to FCRDI and partner farms for image collection	20,000	One-Time	Multiple visits required across 15–20 partner farms for diverse pest image collection
Development Hardware	No additional cost (university lab resources)	0	—	Existing lab GPUs (NVIDIA) used for model training and experimentation

Table 4: Budget Breakdown and Justification

Total Estimated Cost (Year 1): ~LKR 195,000

8. WORK BREAKDOWN STRUCTURE (WBS)

Task Breakdown – Component 2 (Pest Detection – IT22226464)

WBS ID	Task	Duration	Deliverable
2.1	Literature review and research methodology design	Weeks 1–3	Literature review report
2.2	Field visits to FCRDI and partner farms; image collection protocol design	Weeks 3–6	Image collection protocol document
2.3	Collect 2,500+ pest images from FCRDI plots and partner farms	Weeks 4–12	Raw pest image dataset
2.4	Expert annotation and verification by FCRDI entomologists	Weeks 8–14	Annotated bounding box dataset (YOLO + COCO format)
2.5	Data preprocessing: augmentation pipeline, class balance analysis	Weeks 10–16	Preprocessing pipeline code + augmented dataset
2.6	Implement YOLOv8 and Faster R-CNN architectures	Weeks 14–22	Trained baseline detection models

2.7	Small-object optimization: anchor tuning, SAHI integration, FPN enhancement	Weeks 18–24	Optimized small-object detection pipeline
2.8	AutoML hyperparameter search and architecture optimization	Weeks 22–28	Optimal model configuration
2.9	TinyML optimization: INT8 quantization and pruning	Weeks 26–32	Mobile-optimized detection model
2.10	Severity assessment module: pest counting, spatial analysis, heatmap generation	Weeks 28–34	Severity grading pipeline + heatmap module
2.11	IPM recommendation engine development	Weeks 30–34	IPM recommendation module
2.12	Mobile integration and on-device testing	Weeks 32–38	Mobile-deployed pest detection system
2.13	Cross-component API integration (with Components 1, 3, 4)	Weeks 34–40	Integrated API endpoints
2.14	Field validation with entomologists and farmers	Weeks 38–44	Field validation report
2.15	Iterative refinement based on field feedback	Weeks 42–46	Optimized final system
2.16	Documentation and thesis writing	Weeks 44–48	Final dissertation

Table 5: Work Breakdown Structure

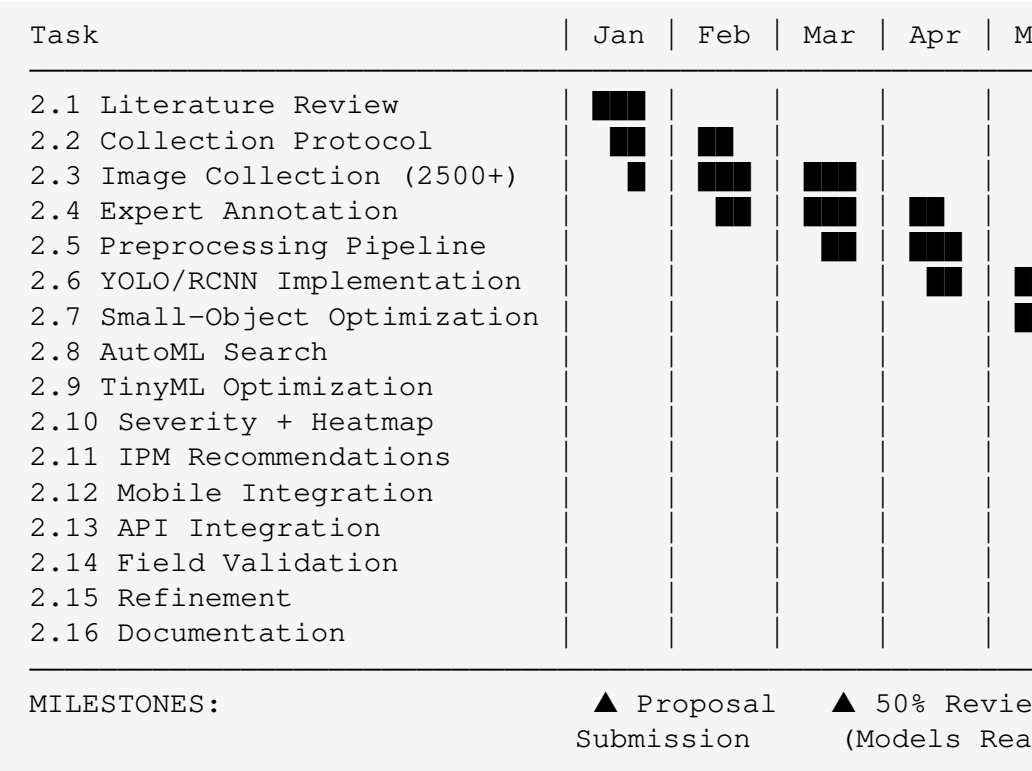
Timeline Alignment with Group

All four members follow a synchronized timeline with shared milestones

at:

- **Week 12:** Data collection complete (all components)
- **Week 24:** Core models trained and individually validated
- **Week 34:** Cross-component integration complete
- **Week 42:** User testing complete
- **Week 48:** Final submission

9. GANTT CHART



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11. APPENDICES

Appendix A: Thrips Infestation Severity Classification Guide

Severity Level	Detection Criteria	Visual Symptoms	Recommended Action
Healthy	No thrips detected; no visible feeding damage	Uniform green coloration; vigorous growth	Continue regular monitoring; maintain preventive measures

Low	1–5 thrips per leaf/ image; minimal feeding marks	Few silvery-white streaks on leaf tips; minor discoloration	Monitor closely; apply neem-based bio-pesticide as preventive; increase sticky trap density
Medium	6–15 thrips per leaf/ image; moderate feeding damage	Widespread silvery streaks; leaf curling beginning; some tip browning	Apply Imidacloprid 17.8% SL at 5ml/10L or Spinosad 45% SC at 0.5ml/L; remove heavily infested leaves
High	>15 thrips per leaf/ image; severe feeding damage	Extensive leaf silvering and curling; necrotic leaf tips; stunted growth; potential leaf drop	Apply systemic insecticide (Fipronil 5% SC at 1.5ml/L); consider foliar spray rotation; implement field sanitation; plan next-season prevention with resistant varieties

Appendix B: Image Collection and Annotation Protocol

Equipment:

- Smartphone camera (minimum 12 MP, autofocus enabled)
- Hand lens (10× magnification) for thrips identification verification
- White paper sheet for contrasting background captures
- Centimeter ruler for scale reference

Collection Procedure:

1. Select representative plants from designated FCRDI plots and partner farms
2. Photograph leaves at 15–30 cm distance showing pest presence or damage symptoms
3. Capture close-up images (<10 cm) of thrips on leaves where visible

4. Include images under natural field conditions (complex backgrounds, variable lighting)
5. Capture images at different times of day (8 AM–4 PM) for lighting variation
6. Record metadata: date, time, plot ID, growth stage, recent pest control activities, weather conditions
7. Capture minimum 5 images per plant from different leaves and angles
8. Use hand lens to verify thrips identity before labeling

Annotation Protocol:

- Use CVAT or Labellmg for bounding box annotation
- Draw tight bounding boxes around individual thrips insects
- Draw separate bounding boxes around visible damage areas (silvery streaks)
- Label classes: `thrips_insect`, `thrips_damage`, `healthy`
- Export in both YOLO format and COCO JSON format

Quality Control:

- Discard blurred, overexposed, or underexposed images
- Minimum resolution: 1024 × 768 pixels
- Review 10% of annotations with entomologist within 48 hours of annotation
- Compute inter-annotator IoU agreement on 10% sample

Appendix C: Model Architecture Comparison Summary

Architecture	Type	Parameters	Inference Speed	mAP (COCO)	Strengths for Pest Detection
YOLOv8-Nano	Single-stage	3.2M	Very Fast	37.3%	Ultra-lightweight; real-time mobile inference

YOLOv8-Small	Single-stage	11.2M	Fast	44.9%	Good accuracy-speed tradeoff; anchor-free
YOLOv8-Medium	Single-stage	25.9M	Moderate	50.2%	Higher accuracy; suitable for fine-grained detection
Faster R-CNN (R50-FPN)	Two-stage	41.1M	Slow	40.2%	Superior localization; better for small objects
Faster R-CNN (R101-FPN)	Two-stage	60.1M	Very Slow	42.0%	Highest accuracy; strong feature extraction
RetinaNet (R50-FPN)	Single-stage	36.3M	Moderate	38.7%	Focal loss for class imbalance; robust

Appendix D: Risk Analysis Matrix

Risk	Probability	Impact	Mitigation
Insufficient thrips image diversity	Medium	High	Collect across multiple growth stages, seasons, and farms; aggressive data augmentation; copy-paste augmentation

Thrips too small to detect at standard resolution	High	High	Use high-resolution input (1280×1280); implement SAHI sliced inference; optimize anchor boxes via AutoML
Class imbalance (pest instances rare vs. background)	High	Medium	Focal loss, copy-paste augmentation, oversampling pest-heavy images, weighted sampling
False positives from similar-looking debris	Medium	Medium	Include hard negative mining; train with diverse background samples; post-processing NMS tuning
mAP degradation after INT8 quantization	Low	High	Quantization-aware training (QAT) if post-training quantization exceeds 3% mAP loss
Field images differ from training data distribution	High	High	Include field-captured images from diverse farms in training; domain adaptation techniques
Seasonal variation in thrips population and appearance	Medium	Medium	Collect images across Maha and Yala seasons; include environmental metadata
Confusion between thrips damage and nutrient deficiency symptoms	Medium	High	Cross-reference with Component 1 and 4 outputs; multi-modal assessment

Appendix E: Ethical Clearance Documents

(To be attached)

- Informed consent form (Sinhala version) for on-farm photography
- Data management plan
- Confidentiality agreement template
- Ethics committee application form (submitted January 2026)